Econ 2 - Lecture 11 - 2/18/26

Short Week: Lecture Quiz #6 Released today, due 2/23

Coffee Hours tomorrow @ 2-3PM

Today: Fiscal Policy = Chapters 4.3 + 5.1

Next Week: National Debt!

↳ Discussion Activity = Gamification

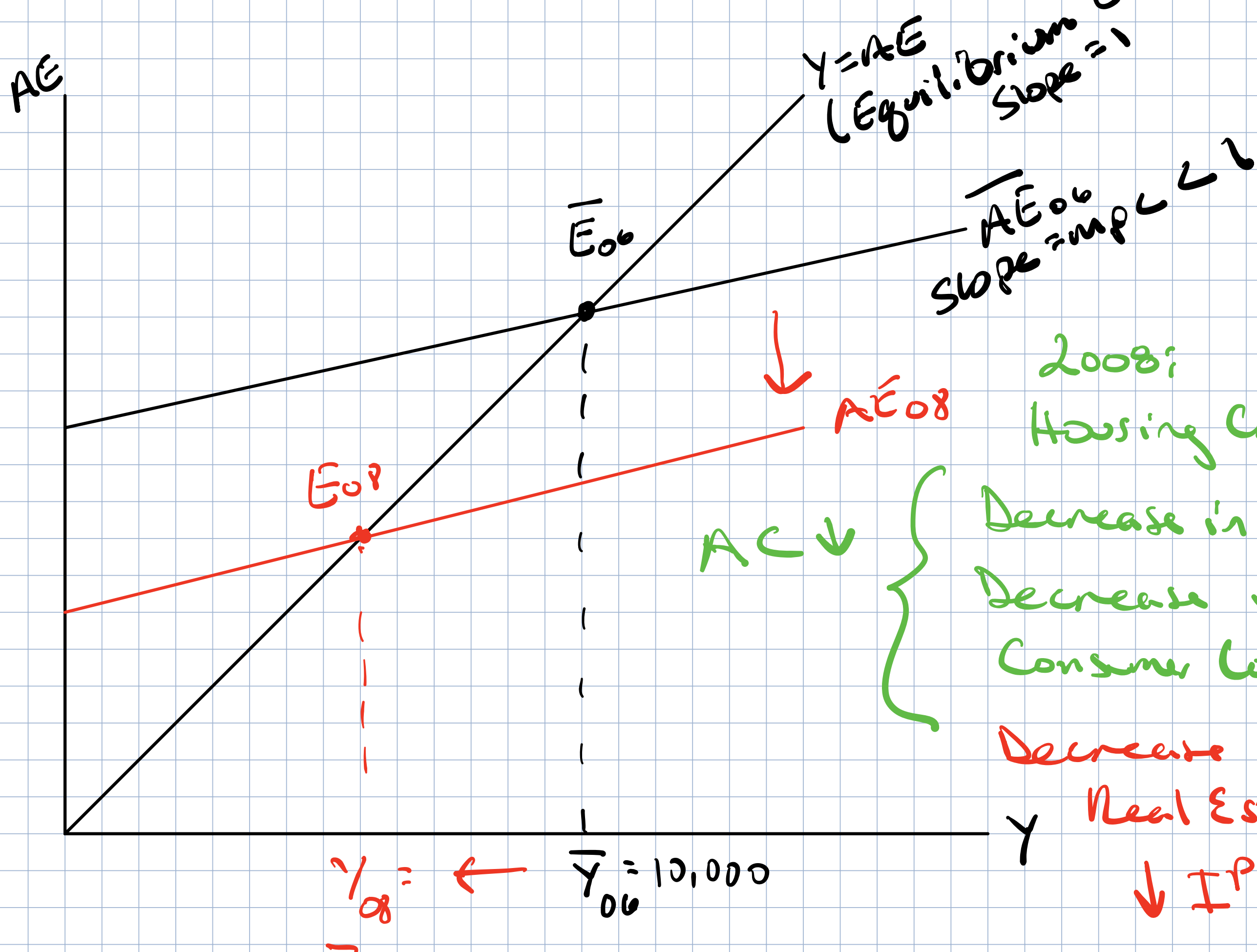
↳ Website → Games Home → Net ID + passcode

Canvas → Grades

Last class: Found where Y = AE! Finally!

Goal of Course: Full-Employment level of $Y = \bar{Y}$ (Y-bar)

→ 0% cyclical UE ⇒ $\bar{Y} = AE$




2008: Housing Crash

AC ↓ { Decrease in Wealth, Decrease in Consumer Confidence }

Decrease in Real Estate

↓ IP

Cyclical Unemployment > 0%

Goal of Policy Makers → get back to $\bar{Y}$

How do we get $AE_{op}$ back to $\overline{AE}$?

Ask for help: Rich, generous individual → Warren Buffett

Need $Y$ to increase by 2500 (7,500 → 10,000)

MPC = 0.8 → We will ask Buffett for...

Setting: $\Delta Y = 2500$, $mpc = 0.8$

Round 1: Ask Warren to buy \$500 in machines/capital

$\Delta I^P = 500$ → $\Delta$ in Income = 500 → $\underline{\Delta Y = 500}$

Marcus increase income

Buy — Save

Car:
$mpc \times 500$
$= 0.8 \times 500$
$= \$400$

Save:
$(1 - mpc) \times 500$
$= 0.2 \times 500$
$= \$100$

$\underline{\Delta Y = 400}$

↓

Chris $\Delta Y = \$400$

Buy — Save

Coffee:
$mpc \times 400$
$= 0.8 \times 400$
$= 320$

Save:
$(1 - mpc) \times 400$
$= 0.2 \times 400$
$= \$80$

$\underline{\Delta Y = 320}$

↓

Neha $\Delta Y = 320$

After 3 Rounds
$\Delta Y = 500 + 400 + 320$
$= 1220$

After ~50 rounds →
$\Delta Y = 0.01$

Total Change in Y after 50 rounds = $\underline{2500}$

$$\underbrace{\text{Total Change in } Y}_{2500} = \underbrace{\text{Initial Change in } I^P}_{500} \times \underbrace{\text{Expenditure Multiplier}}_{?}$$

? = 5 = function of mpc

Expenditure Multiplier =
Number of times each initial dollar change in spending is multiplied through economy

$$\text{Exp. Multiplier} = \frac{1}{1 - mpc}$$

If $mpc = 0.8$, $\text{Exp. Multiplier} = \frac{1}{1-0.8} = \frac{1}{0.2} = 5$

$$\Delta Y = \frac{1}{1-mpc}\Delta I^P = 2500 = \frac{1}{1-0.8}\Delta I^P = 5 \times \Delta I^P$$

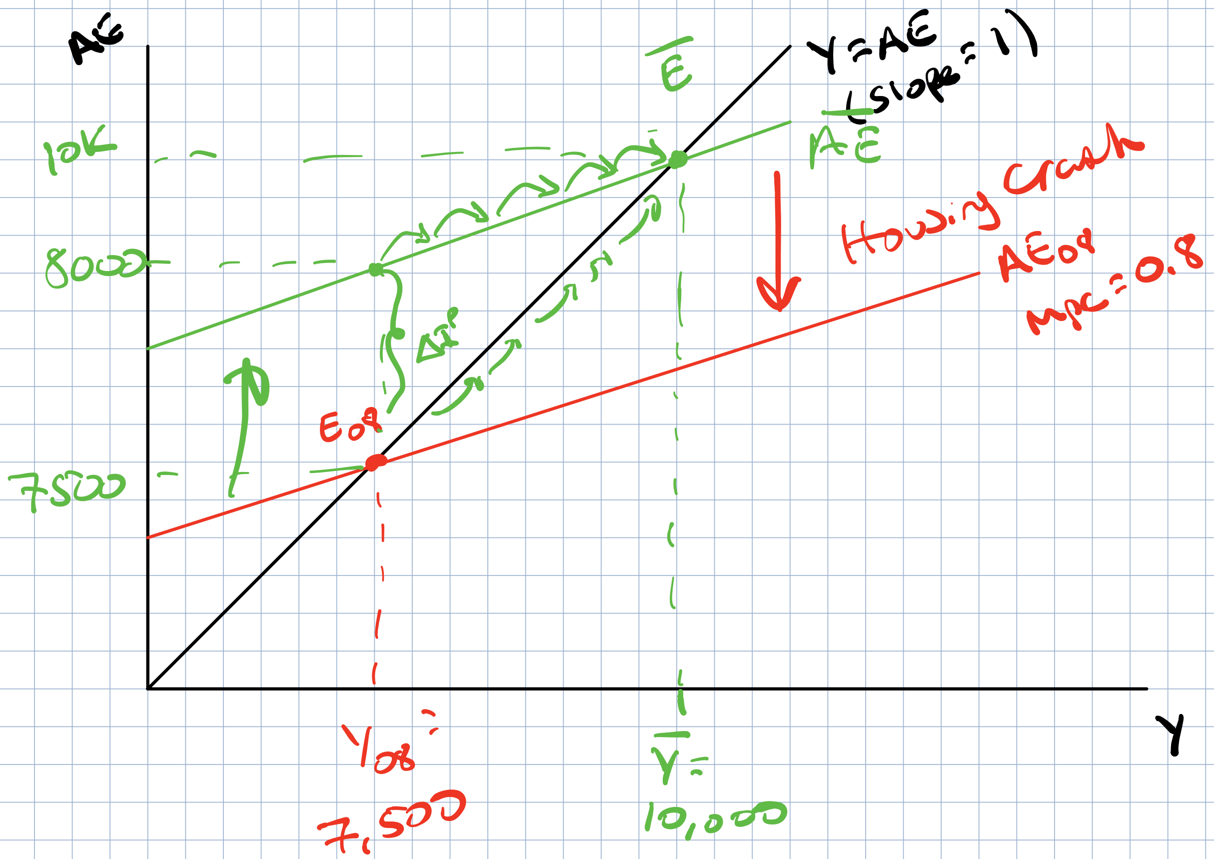
$$\Delta I^P = 500$$

If $mpc = 0.5$ → $\text{Expenditure Multiplier} = \frac{1}{1-0.5} = \frac{1}{0.5} = 2$

$\Delta Y = 2 \times 500 = \$1000$

Work Backwards → Show graphically → multiplier

$Y_{08} = 7{,}500$, $mpc = 0.8$, $\bar{Y} = 10{,}000$




$\Delta I^P = 500$ → in 2008

Housing Crash → $\Delta Y = -2500$

→ $\Delta C < 0$

$\Delta I^P < 0$

Policy Response → Lift up AE → Back to $\bar{AE}$

Expenditure Multiplier application

$\Delta AC, \Delta G, \Delta NX, \Delta I^P \rightarrow \Delta Y = \frac{1}{1 - mpc}\begin{bmatrix} \Delta AC, \\ \Delta I^P, \\ \Delta G, \\ \Delta NX \end{bmatrix}$

$MPC = 0.75$, Expenditure Multiplier $= \frac{1}{1-0.75} = 4$

Housing Crash Reduces autonomous cons. (AC) by $15,000 → total $\Delta Y = ?$

$\Delta AC = -15,000$, $\Delta Y = \frac{1}{1-0.75} \times (-15,000)$

$\Delta Y = 4 \times -15000 = -60,000$

Final Multiplier Analysis

+ 740

C+740

C+840

| Real GDP (Y) | Consumption (C) | + Planned Investment ($I^P$) | + Government Expenditures (G) | | + Net Exports (NX) | = Aggregate Expenditures (AE) | |
|---|---|---|---|---|---|---|---|
| | | | G | G' | | | |
| 5000 | 4560 | 500 | 200 | 300 | 40 | 5300 | 5400 |
| 6000 | 5460 | 500 | 200 | 300 | 40 | 6200 | 6300 |
| 7000 | 6360 | 500 | 200 | 300 | 40 | 7100 | 7200 |
| 8000 | 7260 | 500 | 200 | 300 | 40 | 8000 | 8100 |
| 9000 | 8160 | 500 | 200 | 300 | 40 | 8900 | 9000 |
| 10000 | 9060 | 500 | 200 | 300 | 40 | 9800 | 9900 |

Find $Y = AE$? $MPC = \frac{\Delta C}{\Delta Y} = \frac{5460 - 4560}{6000 - 5000} = \frac{900}{1000} = 0.90$

$Y = AE$ at

→ $Y = 8000$, $AE = 8000$

→ $\bar{Y} = 9000$ → $\Delta G = 100$ → $G' = G + \Delta G = 200 + 100 = 300$

$\Delta Y = 1000 = \frac{1}{1-MPC} \Delta G = \frac{1}{1-0.9}$ $\Delta G = 10 \times G$

$\Delta G = \frac{1000}{10}$

Recap: Find short-run equilibrium ($Y = AE$)

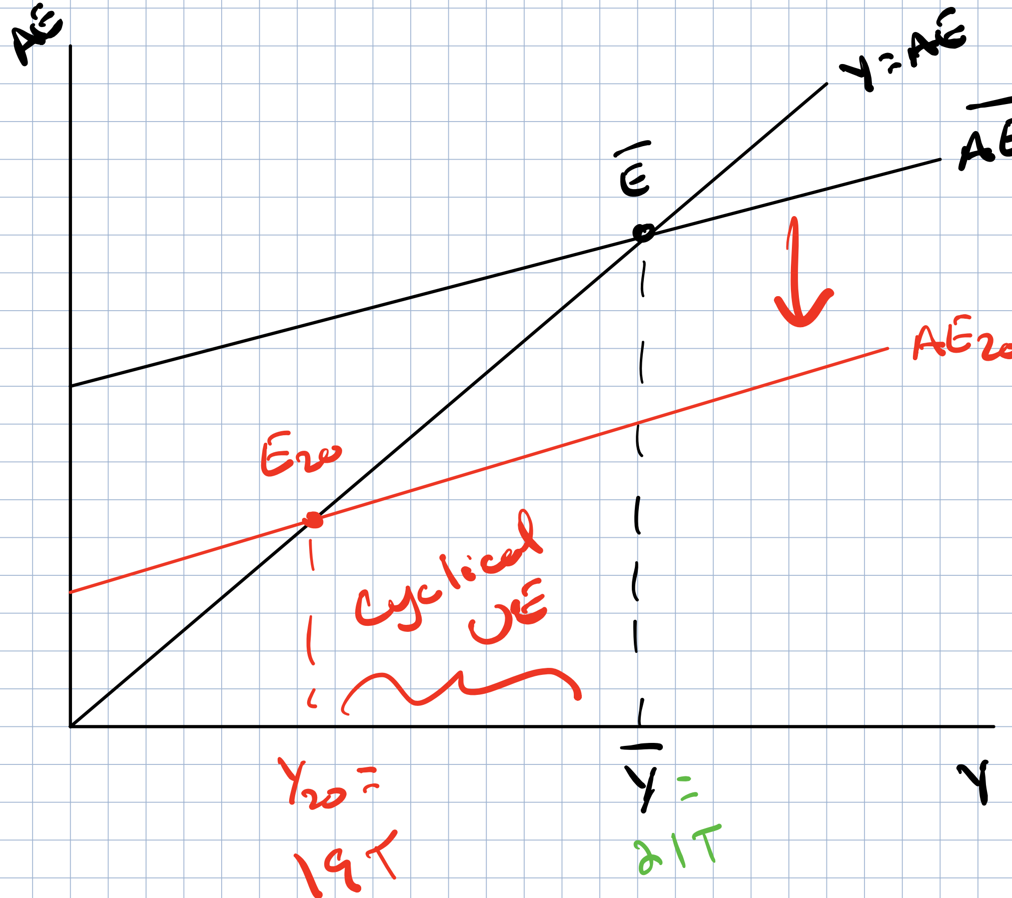
Table, graph, story, etc

Incorporate economic shocks into model

Housing Crash $\rightarrow \downarrow AC, \downarrow I^P \Rightarrow \Delta Y = \frac{1}{1-mpc}(\Delta AC + \Delta I^P)$

More Recent: COVID-19 Pandemic

In 2019: Full-Employment $\bar{Y} \Rightarrow$ UE = 3.5% in $\underbrace{\text{Dec. 2019}}_{\text{Real GDP = 21T}}$




In 2020:
Lockdown
$\downarrow AC, \downarrow I^P$
$\downarrow AE$

UE Rate April 2020:
14.7%!

Real GDP = 19T